

Asymptotic Space Complexity

Improve the space complexity

- Reverse an array

SATISH becomes HSITAS

Pop quiz

An algorithm to reverse a character array of n letters using a temporary array of n letters has

(choose the option from the list below)

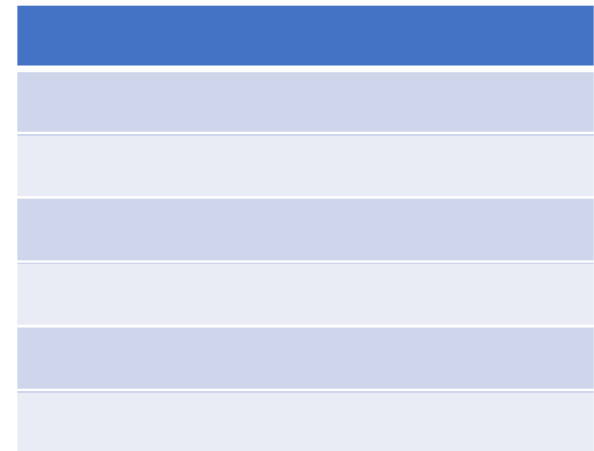
As an example, algorithm to reverse the array converts **RAINMAN** to **NAMNIAR**.

- ☐ time complexity of $O(N)$ and space complexity of $O(1)$
- ☐ time complexity of $O(N)$ and space complexity of $O(N)$
- ☐ time complexity of $O(1)$ and space complexity of $O(N)$
- ☐ time complexity of $O(N)$ and space complexity of $O(1)$

Space complexity of recursive programs

Recursive programs

```
int sum (int n)
{
    if(n <= 0 )
        return 0;
    return n + sum(n-1);
}
```



Space complexity question

```
void traverse( int a )
{
    if(a == 0)
        return;
    else
    {
        a = a - 1;
        traverse(a);
    }
}

main()
{
    traverse(100);
}
```

Options:

$O(100)$

$O(a)$

$O(1)$

$O(a * a)$